CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (EC) Examination

May 2014

EC308 Antenna & Wave Propagation

Date: 06.05.2014, Tuesday | **Time: 10:00 a.m. To 01:00 p.m.** | **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Define the following terms. [07]

1) HPBW
2) Directivity
3) Poynting Vector
4) Isotropic Point Source
5) Radiation Intensity
6) Pitch Angle
7) Effective Aperture

Q - 2 (a) Define Pattern Multiplication principle. Using the pattern multiplication principle, explain the radiation pattern of 4 isotropic elements fed in phase, spaced $\lambda/2$. [07]

(b) Starting from retarded currents, derive equations for the E components for a short dipole antenna if spherical system is defined in terms of r, Θ,ϕ [07]

OR

Q - 2 (a) Design a broadside Dolph Tschebycheff array of 10 elements with spacing d between the elements and with a major to minor lobe ratio of 26dB. Find the excitation coefficients and array factor. [07]

(b) If a short antenna is operated on 1 MH_z frequency having length of 5 cm, the current flowing through antenna is 5 mA, find the power radiated by antenna and radiation resistance. [07]

Q - 3 (a) Prove that radiation resistance of a small loop antenna depends on area of loop. [04]

(b) Define: Axial ratio for helical antenna. Also derive its equation. [06]

(c) Derive the impedance for 2 wire folded dipole. [04]

OR

Q - 3 (a) Compare the far field equations of small loop with small dipole. [04]

(b) Explain designing parameters of helical antenna. [06]

(c) Design a Rhombic antenna operated on 1 MHz with 30^0 elevation angle. Find the Electric field intensity for designed Antenna. [04]

SECTION – II

Q - 4 (a) Explain Fermat's principle for equality of path length in lens antenna. [04]

(b) Compare V Antenna and Rhombic Antenna. [03]

Q - 5 (a) Discuss significance of complementary antenna with the help of Babinet's principle. [04]

(b) Explain the experimental setup for the measurement of Radiation pattern of antenna [06]

(c) A paraboloid antenna with a circular aperture is to have a power gain of 1000 at λ=10cm. find the diameter of the mouth and HPBW of the antenna. [04]

OR

Q - 5 (a) Explain types of horn antenna [04]

(b) Explain the method for the impedance measurement. [06]

(c) Discuss the offset paraboloid reflector. [04]

Q - 6 (a) In satellite communication systems, the height of the satellite is 36,000 km above the earth, and operated at 4000MHz. the gain of the transmitting antenna is 20 dB and that of receiving antenna is 40 dB. Find (i) the free space transmission loss (ii) the power received when the power transmitted is 200 W. Assume free space conditions. [06]

(b) Design a patch antenna operated on 10GHz using a substrate with dielectric constant of 2.2, height of substrate h=0.1588 cm. [04]

(c) Design 3 Element Yagi-Uda Antenna operating on 100MHz. [04]

OR

Q - 6 (a) Define (a)skip distance (b) duct propagation (c) MUF [06]

(b) Explain different types of feeding methods in micro strip antenna. [04]

(c) Explain the Design Steps of log periodic Antenna with help of Example. [04]

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